



A dual-branch RGB-T salient object detection via spatial-frequency integration

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Abstract

RGB-T salient object detection aims to fuse information from visible and thermal infrared modalities to accurately localize salient regions in complex scenes. However, existing methods primarily rely on spatial-domain features for cross-modal fusion and fail to fully exploit the potential of frequency-domain information. Additionally, inherent differences in spatial structures and frequency distributions often constrain cross-domain feature fusion, leading to suboptimal fusion results. To address these issues, we propose a dual-branch spatial-frequency integration network (SFNet), designed to effectively achieve complementary fusion and cross-domain alignment of bimodal information. Specifically, we introduce a spatial-domain cross-modal fusion module and a frequency-domain cross-modal perception module, which capture long-range dependencies and global structural information, respectively. Furthermore, to overcome semantic inconsistencies between cross-domain features, we propose a dual-domain semantic interaction module that enhances fusion through semantic alignment and feature interaction. Additionally, a feature decoupling module is designed to extract both the main and edge details of the fused features, further improving localization accuracy and boundary clarity through a custom loss function. Experimental results show that SFNet outperforms state-of-the-art methods across multiple benchmark datasets, demonstrating its superior robustness and broad generalization capability. The code is available at <https://github.com/icehire/SFNet>.

Keywords RGB-T salient object detection · Cross-modal feature fusion · Cross-domain feature fusion · Feature decoupling

1 Introduction

Salient object detection (SOD) seeks to emulate human visual mechanisms by automatically identifying and segmenting the most visually prominent objects or regions within a variety of scenes. As a cornerstone task in computer vision, SOD

has found widespread application in domains such as image segmentation [1], image compression [2], object detection [3–6], object tracking [7–9], 3D reconstruction [10, 11] and remote sensing image processing [12, 13].

Early research on salient object detection [14–16] predominantly relied on RGB images. These traditional methods struggle to accurately detect objects in complex scenarios, such as dark environments, object occlusions, and similar foreground–background coloration. To address these challenges, researchers have increasingly incorporated multimodal information, such as thermal imaging [17] and depth data [18], to improve scene understanding and enhance detection accuracy. Depth images depict the three-dimensional structure and spatial relationships of objects within a scene. Thermal infrared images, which reflect thermal radiation differences among objects, are particularly effective at characterizing object information and demonstrate strong adaptability in adverse weather or low-light environments. Significant progress has been made in salient object detection using both RGB and thermal infrared images [19].

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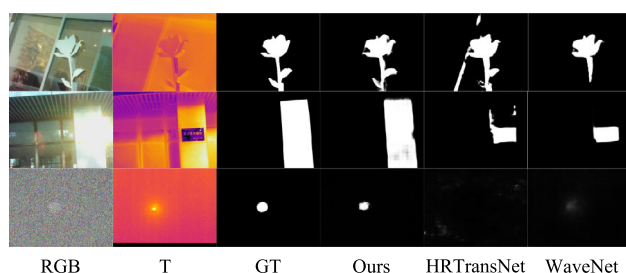


Fig. 1 Visual comparisons between the proposed SFINet and representative RGB-T SOD models, including HRTransNet [26] and WaveNet[27] in complex scenes

In recent years, the Transformer model [20] has effectively captured the global dependencies between features through self-attention mechanisms, improving the processing of high-level semantic features and demonstrating outstanding performance in various computer vision tasks [21–23]. The performance of RGB-T SOD models based on Transformer has also been correspondingly enhanced [24–26]. However, such models primarily focus on spatial-domain feature processing, failing to fully exploit the inherent frequency-domain information in images. To address this limitation, a recent study [27] introduced the discrete wavelet transform (DWT) into the SOD task, facilitating cross-modal information fusion in the frequency domain. While frequency-domain learning can expand the receptive field and effectively capture rich global information, relying solely on frequency features may undermine the model's semantic understanding of complex scenes and its ability to precisely perceive spatial locations. As illustrated in Fig. 1, in challenging scenarios such as irregular objects (1st row), strong illumination interference (2nd row), and image information degradation or loss (3rd row), methods like HRTransNet [26] and WaveNet [27] exhibit poor performance in object structure inference and boundary localization, suffering from substantial missed detections or mis-segmentations. Therefore, enhancing model robustness depends on the effective integration of frequency-domain global context with high-level semantic fusion, facilitating cross-domain complementarity and synergy.

To address the aforementioned challenges, this paper proposes the following solutions. Firstly, we propose a spatial-domain cross-modal fusion attention (SCFA) module and a frequency-domain cross-modal perception attention (FCPA) module, each responsible for effectively fusing bimodal features in the spatial and frequency domains, respectively. The SCFA module, grounded in the attention mechanism, utilizes a parameter-sharing strategy to facilitate hybrid intra-modal and inter-modal interactions. This enhancement improves the model's capacity to capture long-range dependencies, augments its understanding of spatial structures within scenes, and fosters complementary exchange of cross-

modal semantic information. In the frequency domain, the fast Fourier transform (FFT) offers strong global representational capabilities, enabling the decomposition of an image into amplitude and phase components, which effectively represent the shape and edge details of objects. The FCPA module utilizes FFT to decompose and recombine features along both spatial and channel dimensions, thereby enhancing individual features at multiple levels, reducing inter-modal information discrepancies, and enabling deep fusion of cross-modal features. Meanwhile, to overcome the semantic discrepancies between cross-domain features, we propose the dual-domain semantic interaction module (DSIM). This module first uses strip convolutions to map cross-domain features to the same scale and then performs semantic interaction and fusion based on cross-attention.

Furthermore, to fully leverage the synergistic advantages between global features and local details, we propose the design of a feature decoupling module (FDM), which is grounded in a label decoupling strategy. The FDM is composed of two branches: a Body Capture Branch (BCB) and an Edge Detail Capture Branch (EDCB), specifically designed to capture the primary body region and boundary details of the salient object, respectively. These branches are further enhanced by the integration of shallow-level features to bolster their representational capacity. The corresponding body and edge detail labels are derived from the ground truth of the original salient object. Ultimately, the model generates the final prediction map under the joint supervision of the body loss, edge detail loss, and overall saliency loss.

In summary, the main contributions of our work are as follows:

- 1 We introduce an dual-branch spatial-frequency integration network for RGB-T salient object detection, referred to as SFINet, which integrates cross-modal feature fusion across both spatial and frequency domains, demonstrating superior performance in both quantitative evaluations and model comparisons.
- 2 We introduce the SCFA and FCPA as novel approaches to facilitate the fusion of cross-modal features across diverse domains. Meanwhile, we present the DSIM, which aims to establish a representational correlation of identical features across domains, thereby mitigating semantic discrepancies. This approach enables effective and complementary fusion of cross-domain information.
- 3 We design the FDM that comprehensively integrates both global structural features and local detailed features of the target, thereby enhancing the model's capacity to localize and comprehend salient objects. By integrating the body loss and edge detail loss, this module significantly improves the prediction accuracy.

2 Related work

2.1 RGB-T salient object detection

Traditional RGB-T salient object detection methods [28] have primarily relied on handcrafted feature extraction and graph-based association techniques. Wang et al. [17] pioneered the multimodal SOD task, which integrates RGB and thermal infrared images, by constructing the first RGB-T dataset, VT821. With the advancement of deep neural networks, an increasing amount of research has focused on exploiting deep learning models to uncover complementary information across modalities, thereby improving detection accuracy. Zhou et al. [29] addressed the scale discrepancies between dual-modal features by designing a consistency fusion network, which improves object extraction accuracy by integrating complementary information. Zhou et al. [30], focusing on model lightweight design, proposed a network architecture that incorporates knowledge distillation and transfer learning strategies to strengthen multimodal feature fusion capabilities. Xu et al. [31] aimed at minimizing inter-modal discrepancies while preserving key features of each modality. They introduced a fully fused module and a gating mechanism in the decoding stage, enabling selective filtering and fusion of features, significantly enhancing detection performance. Additionally, Tang et al. [32] proposed introducing a complementary modality stream into a dual-stream network, thereby constructing a tri-stream interactive architecture that effectively mitigates modality discrepancies and substantially improves detection quality.

In recent years, the Transformer architecture has been widely adopted in various computer vision tasks, including panoptic segmentation [33], object detection [34], and image classification [35]. Liu et al. [25] employed a dual-stream Swin Transformer encoder to extract bimodal features, significantly improving detection accuracy by integrating spatial alignment and channel calibration techniques. Luo et al. [36] constructed a dual-stream encoder based on Transformers, utilizing attention mechanisms to fuse complementary information, enhancing background context utilization and effectively reducing noise during the fusion process. Zhou et al. [37] introduced a position-aware relational learning network, which strengthens the relationships between isolated boundary pixels and other confident pixels, enabling multimodal feature decoding based on Transformers. Pang et al. [38] proposed a cross-modal view-mixing Transformer, utilizing a view-mixing attention mechanism to propagate context and update sequences across modalities.

Contrary to the aforementioned methods, we utilize a convolutional backbone network to extract multi-scale features. Furthermore, we design distinct attention interaction mechanisms that account for modality and domain differences,

thereby facilitating semantic alignment and ensuring feature complementarity.

2.2 Networks based on frequency feature learning

Frequency-domain analysis serves as a crucial bridge between pixel-level representations and high-level semantic features, effectively addressing challenges such as boundary blurring and noise interference in the spatial domain. It has been widely applied in tasks such as image segmentation [39], image fusion [40], image denoising [41], and image restoration [42]. Common methods for transforming images into the frequency domain include the FFT, wavelet transform [43, 44], and discrete cosine transform (DCT) [45]. For example, Tatsunami et al. [46] designed the DFFormer and CDDFormer encoders specifically for processing frequency-domain features, demonstrating exceptional performance in downstream tasks. Duan et al. [47] enhanced their model's ability to capture both global and local features through Fourier transforms, resulting in precise segmentation of small objects in remote sensing images. Zhou et al. [27] utilized a multilayer perceptron (MLP) based on wavelet transforms for bimodal feature extraction, integrating knowledge distillation and cross-modal fusion strategies. This approach not only maintains a lightweight model but also significantly improves the performance of RGB-T salient object detection. Zhong et al. [48] employed the discrete cosine transform to thoroughly extract frequency-domain features of targets, achieving performance that surpasses human vision. Zhou et al. [49] leveraged the separability and complementarity of frequency-aware features to fuse bimodal features and generate high-frequency and low-frequency cues. This approach effectively mitigates issues of boundary blurring and single representation in RGB-T salient object detection. Given the complementary advantages of spatial-domain processing and frequency-domain analysis, combining these two approaches enhances the model's ability to understand and interpret complex image scenes. By exploiting the strengths of spatial-domain attention and frequency-domain Fourier analysis, this integration facilitates cross-domain interaction and semantic alignment of features, improving the model's comprehension of complex scenes.

3 Proposed method

3.1 Overall architecture

The overall framework of the proposed SFNet is illustrated in Fig. 2. Within this framework, ConvNeXt-B is employed as the backbone network for extracting features from RGB and thermal infrared images separately. ConvNeXt preserves the advantages of convolutional neural networks, such as trans-

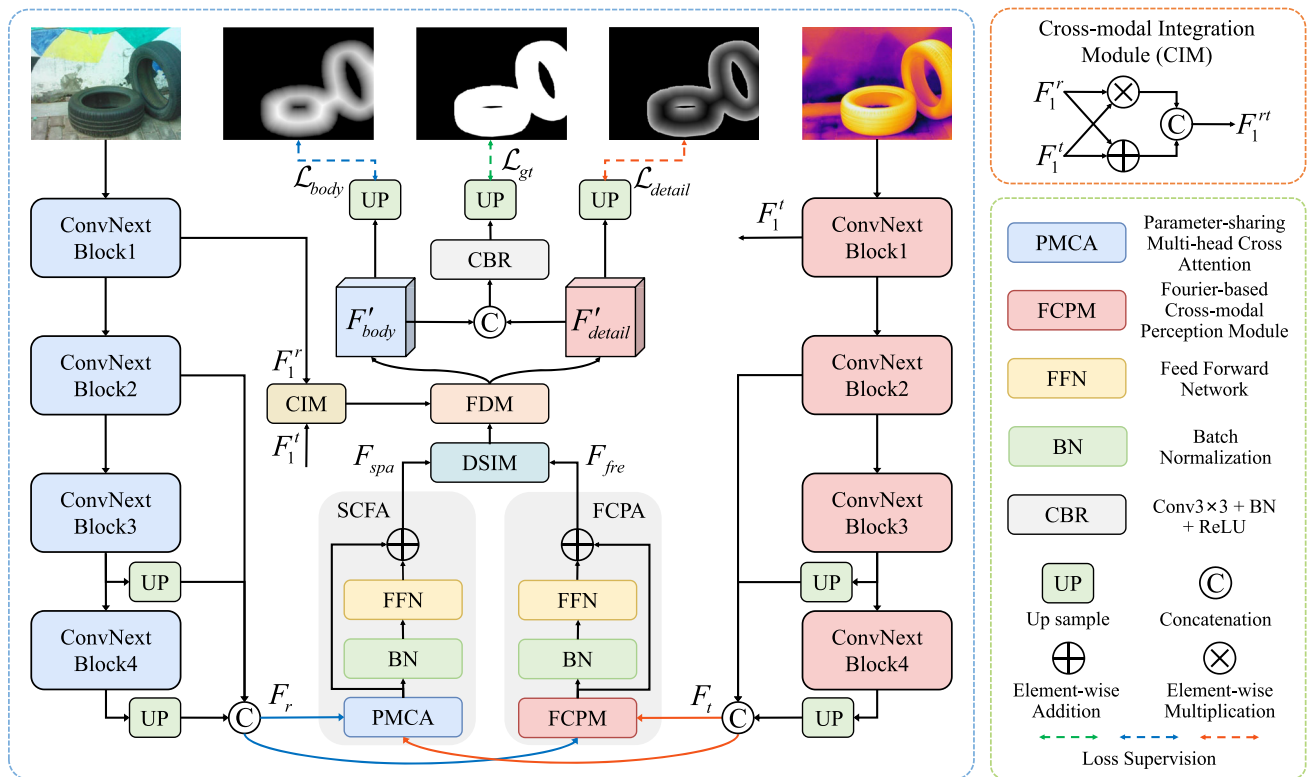


Fig. 2 The overall architecture of our proposed SFINet. RGB and thermal images first undergo feature extraction through a dual-stream ConvNeXt encoder. SCFA and FCPA perform spatial-domain and frequency-domain fusion of high-level features from the dual modalities, respectively. Subsequently, DSIM aligns the cross-domain features

semantically and achieves complementary fusion. Finally, FDM decouples the fused features, separating them into body features and edge detail features, and incorporates low-level features to enhance representational capacity

lation invariance and low data dependency, while integrating design principles from Transformers. This integration significantly improves both feature extraction capabilities and computational efficiency, leading to superior performance in multiple downstream tasks.

Firstly, the RGB images and thermal infrared images are input into dual-stream ConvNeXt networks to extract multi-scale features $\{r_i\}_{i=1}^4$ and $\{t_i\}_{i=1}^4$, respectively. Concurrently, a 1×1 convolution is applied to uniformly reduce the number of channels in the multi-scale features to 96. Then, the first-layer bimodal features are then passed into the Cross-modal Integration Module (CIM). Within this module, the RGB features and thermal infrared features undergo element-wise addition and multiplication operations. The results are then concatenated along the channel dimension, and feature fusion is performed through convolution. The processing procedure of $CIM(\cdot)$ can be expressed as:

$$F_1^{rt} = \text{Cat}((r_1 + t_1), (r_1 \otimes t_1)), \quad (1)$$

where $+$ and $\otimes$ denote element-wise addition and multiplication operations, respectively. $\text{Cat}(\cdot)$ represents channel concatenation, and $F_1^{rt} \in \mathbb{R}^{2C \times h \times w}$ denotes the result of the

first-layer bimodal feature fusion. Here, C , h , and w are all equal to 96.

Subsequently, the final three feature layers from each modality, which contain rich high-level semantic information, are concatenated along the channel dimension. This operation can be expressed as:

$$\begin{cases} F_r = \text{Cat}(r_2, \text{Up}(r_3), \text{Up}(r_4)), \\ F_t = \text{Cat}(t_2, \text{Up}(t_3), \text{Up}(t_4)), \end{cases} \quad (2)$$

where $\text{Up}(\cdot)$ denotes the upsampling operation. $F_r, F_t \in \mathbb{R}^{3C \times H \times W}$ represent the RGB features and thermal infrared features, respectively, where H and W correspond to the height and width of r_2 and t_2 . Subsequently, the RGB features F_r and thermal infrared features F_t are input into the SCFA and FCPA modules, respectively, to facilitate feature fusion in the spatial and frequency domains. This fusion process enables the effective complementarity between the cross-domain features. Following this, the DSIM is utilized to semantically align the features in both the spatial and frequency domains. Subsequently, the FDM separates the bi-domain fused features into body and edge detail features. Under the supervision of ground-truth data, these features

are employed to guide the prediction of the main body and edge regions, respectively. Finally, the body and edge detail features are concatenated along the channel dimension. The resulting feature map is then processed through a Convolution-BatchNorm-ReLU (CBR) block and subsequently upsampled to generate the final prediction.

3.2 Dual-domain cross-modal fusion

RGB images encapsulate a rich array of color, texture, and foreground-background information, whereas thermal infrared images capture temperature variations among objects. To fully leverage the advantages of both modalities and achieve complementary and collaborative representations, this paper proposes a parallel dual-domain fusion strategy that integrates spatial and frequency-domain perspectives, aiming to thoroughly explore the similarities and complementary relationships between the modalities. Specifically, we propose a spatial-domain cross-modal fusion attention module and a frequency-domain cross-modal perception attention module, which facilitate adaptive feature fusion at the spatial structure and spectral characteristic levels, respectively. The proposed method offers a novel perspective on cross-modal feature fusion and significantly enhances the model's ability to represent and integrate multimodal, multi-domain features.

3.2.1 Spatial-domain cross-modal fusion attention

Effectively addressing the physical disparities between modalities and achieving comprehensive fusion of critical information from diverse images is pivotal for improving the performance of saliency detection. Current methods for high-level semantic fusion predominantly rely on cross-attention mechanisms. However, these approaches generally facilitate only shallow feature vector interactions, and their fusion effectiveness is often limited by the resolution of feature maps. As a result, it becomes challenging to achieve adequate cross-modal interaction, particularly in complex scenes. In contrast, the proposed SCFA module introduces intra-modal vector interactions, cross-modal parameter-sharing, and a hybrid interaction strategy. This approach thoroughly leverages long-range dependencies between bimodal features, enabling both self-enhancement and mutual enhancement of features. Consequently, the efficiency of complementary information utilization is improved, thereby increasing the model's robustness and discriminative capacity in complex environments.

The structure of the SCFA module is shown in Fig. 2, comprising three key components: a parameter-sharing multi-head cross-attention $PMCA(\cdot)$, batch normalization layer $BN(\cdot)$, the feed-forward network $FFN(\cdot)$ and residual connections. The entire process can be expressed as:

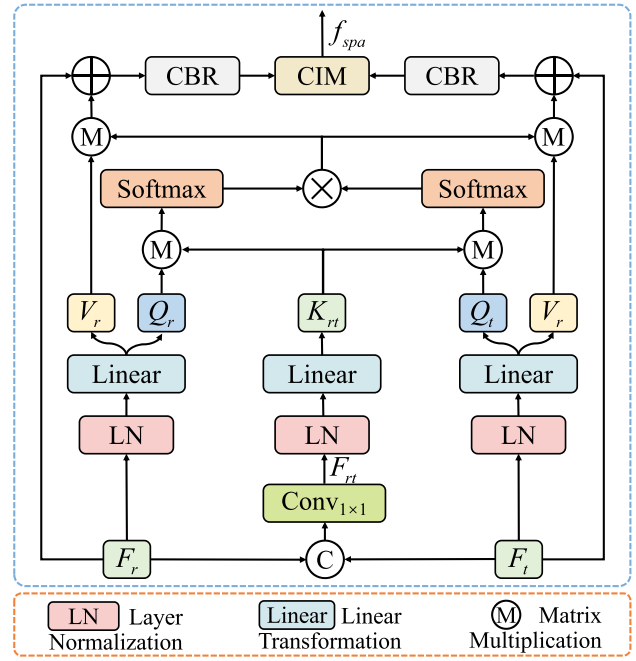


Fig. 3 Framework of proposed parameter-sharing multi-head cross-attention (PMCA)

$$f_{spa} = PMCA(F_r, F_t), \quad (3)$$

$$F_{spa} = FFN(BN(f_{spa})) + f_{spa}, \quad (4)$$

where F_{spa} represents the spatial-domain cross-modal fused features.

In the $PMCA(\cdot)$ module, to achieve parameter sharing between modalities, we first concatenate the features F_r and F_t along the channel dimension. Then, we use a convolution operation to fuse the concatenated result and reduce its channel number by half, thereby obtaining the shared feature F_{rt} . Next, we flatten the last two dimensions of the F_r, F_t, F_{rt} , adjust their sizes to $F_r, F_t, F_{rt} \in \mathbb{R}^{HW \times 3C}$, and use the layer normalization operation $LN(\cdot)$ and linear transformation $Linear(\cdot)$ to process F_r and F_t separately to obtain the single-modality query vectors Q_r and Q_t , as well as the single-modality value vectors V_r and V_t . Simultaneously, we apply the same processing steps to F_{rt} to obtain the parameter-shared key vector K_{rt} . The above process is shown in Fig. 3 and can be formally expressed as:

$$F_{rt} = Conv(Cat(F_r, F_t)), \quad (5)$$

$$K_{rt} = Linear(LN(F_{rt})), \quad (6)$$

$$Q_r = Linear(LN(F_r)), V_r = Linear(LN(F_r)), \quad (7)$$

$$Q_t = Linear(LN(F_t)), V_t = Linear(LN(F_t)). \quad (8)$$

By interacting the parameter-shared key vector K_{rt} with the query vectors from each modality, we achieve both the self-enhancement of intra-modal features and the self-adaptive

Different channels of an image carry distinct perceptual information, with their frequency-domain responses manifesting as different features in the amplitude spectrum. CFFT effectively extracts the frequency information from each channel, separating low-frequency common components from high-frequency differential components. This helps identify cross-modal shared significant structures and boundaries while suppressing modality-specific noise interference.

Thus, we perform channel-wise Fourier transform and decomposition on α_{rs} and α_{ts} , and enhance the representation capability of the two groups of channel-wise Fourier components using the $CLC(\cdot)$ operation. Subsequently, we combine these two groups of channel-wise Fourier components into complex form and execute the inverse Fourier transform to obtain the enhanced amplitude vectors $\tilde{\alpha}_{rs}, \tilde{\alpha}_{ts} \in \mathbb{R}^{3C \times H \times (W/2)}$. Simultaneously, we process the spatial Fourier phase components p_{rs} and p_{ts} using the $CLC(\cdot)$ operation to obtain the enhanced phase vectors $\tilde{p}_{rs}, \tilde{p}_{ts} \in \mathbb{R}^{3C \times H \times (W/2)}$. The entire process can be expressed as:

$$\begin{cases} f_{rc} = CFFT(\alpha_{rs}), & f_{tc} = CFFT(\alpha_{ts}), \\ a_{rc} = Amp(f_{rc}), & p_{rc} = Pha(f_{rc}), \\ a_{tc} = Amp(f_{tc}), & p_{tc} = Pha(f_{tc}), \\ a'_{rs} = Complex(CL C(a_{rc}), CL C(p_{rc})), \\ a'_{ts} = Complex(CL C(a_{tc}), CL C(p_{tc})), \end{cases} \quad (17)$$

$$\begin{cases} \tilde{a}_{rs} = ICFFT(a'_{rs}), & \tilde{a}_{ts} = ICFFT(a'_{ts}), \\ \tilde{p}_{rs} = CL C(p_{rs}), & \tilde{p}_{ts} = CL C(p_{ts}), \end{cases} \quad (18)$$

where $CLC(\cdot)$ denotes a series of operations including two 1×1 convolutional layers and a LeakyReLU activation function.

To establish the relationship between the enhanced amplitude components and the original amplitude components, we perform element-wise multiplication between them and combine the result with the enhanced phase components to form complex numbers. Subsequently, we add the two complex features and use the inverse Fourier transform to convert them back to the spatial domain, thereby completing the fusion of cross-modal information. Additionally, to avoid loss of original information, we also add the input features to the fused results to further enhance the representation capability of the features. The entire process can be expressed as:

$$f'_{rs} = Complex((a_{rs} \otimes \tilde{a}_{rs}), \tilde{p}_{rs}), \quad (19)$$

$$f'_{ts} = Complex((a_{ts} \otimes \tilde{a}_{ts}), \tilde{p}_{ts}), \quad (20)$$

$$f_{fre} = IFFT(f'_{rs} + f'_{ts}) + F_r + F_t. \quad (21)$$

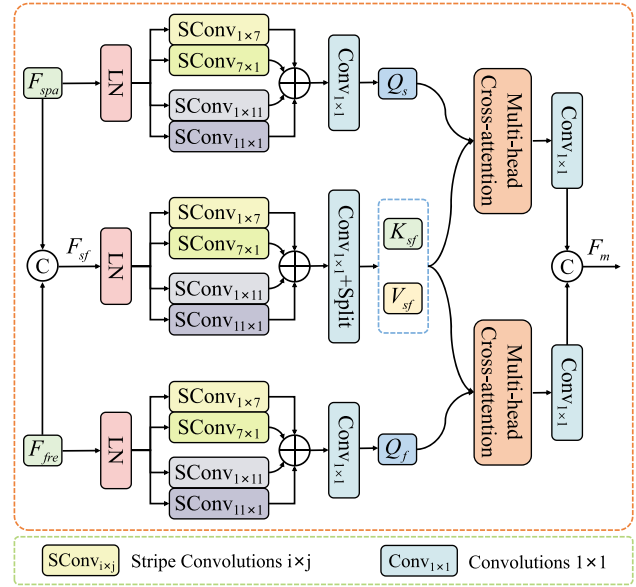


Fig. 5 Framework of proposed dual-domain semantic interaction module (DSIM)

3.3 Dual-domain semantic interaction module

Spatial-domain and frequency-domain processing represent two distinct approaches for image feature analysis. Each domain exhibits unique characteristics and emphasizes different aspects of image features, with the spatial domain focusing on local structures and the frequency domain capturing global patterns, leading to inherent semantic differences. Consequently, effective cross-domain feature fusion requires semantic alignment to ensure both consistency and complementarity in image feature representation. To this end, we propose the DSIM, illustrated in Fig. 5.

To achieve consistency and complementarity among features from different domains, we initially concatenate F_{spa} and F_{fre} along the channel dimension to obtain the complementary fusion feature F_{sf} . This feature will guide the semantic alignment of F_{spa} and F_{fre} through parameter sharing. Subsequently, multi-scale strip convolution is used to map the spatial-domain features, frequency-domain features, and complementary fusion features to a unified scale, thereby reducing their semantic differences. The above process can be expressed as:

$$F_{sf} = Cat(F_{spa}, F_{fre}), \quad (22)$$

$$\begin{cases} F'_{spa} = \delta_7(LN(F_{spa})) + \delta_{11}(LN(F_{spa})), \\ F'_{fre} = \delta_7(LN(F_{fre})) + \delta_{11}(LN(F_{fre})), \\ F'_{sf} = \delta_7(LN(F_{sf})) + \delta_{11}(LN(F_{sf})), \end{cases} \quad (23)$$

where δ_k denotes the $1 \times k$ and $k \times 1$ strip convolution layers. Subsequently, we apply 1×1 convolution operations to the

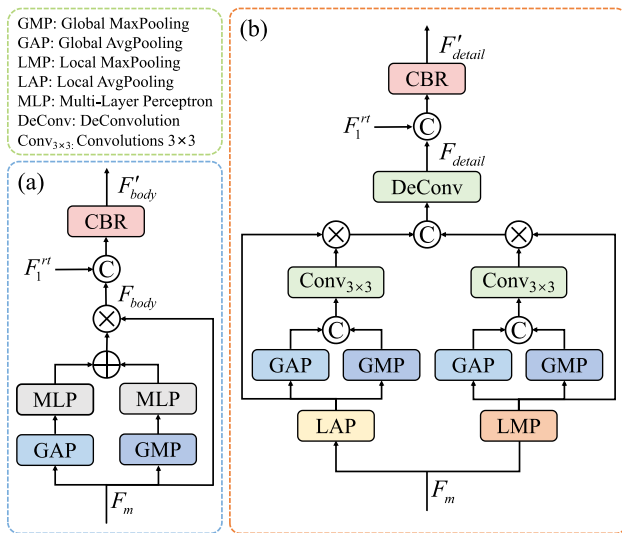


Fig. 6 This figure illustrates the process of the feature decoupling module (FDM). **a** shows the framework for BCB, and **b** depicts the procedure for EDCB

transformed features separately to obtain the query vectors for attention computation from the single modality, and the key and value vectors from the complementary modality:

$$\begin{cases} Q_s = \text{Conv}_{1 \times 1}(F'_{spa}), \\ Q_f = \text{Conv}_{1 \times 1}(F'_{fre}), \\ K_{sf}, V_{sf} = \text{Split}(\text{Conv}_{1 \times 1}(F'_{sf})), \end{cases} \quad (24)$$

where $\text{Split}(\cdot)$ denotes splitting the feature along the channel dimension. Finally, we perform multi-head cross-attention operations to complete semantic alignment:

$$\begin{cases} \text{Attn}_s = \text{Softmax}(Q_s \cdot K_{sf}^T / \sigma) \cdot V_{sf}, \\ \text{Attn}_f = \text{Softmax}(Q_f \cdot K_{sf}^T / \sigma) \cdot V_{sf}, \end{cases} \quad (25)$$

$$F_m = \text{Cat}(\text{Conv}_{1 \times 1}(\text{Attn}_s), \text{Conv}_{1 \times 1}(\text{Attn}_f)). \quad (26)$$

3.4 Feature decoupling module

The label decoupling supervision strategy decomposes the ground-truth labels of salient objects into two components: the main body labels and the detail labels. These components are then used in conjunction with the original labels for supervision during training [50]. Motivated by this approach, we observe that the main body of a target typically demonstrates internal structural consistency, making its features relatively easier to capture. In contrast, fine-grained details, such as textures and edges, are more complex and necessitate modeling semantic similarities within local spatial neighborhoods. Building on this insight, we propose the FDM, as shown in Fig. 6, which disentangles the semantically aligned feature

F_m , thus facilitating more effective subsequent loss supervision. The proposed module consists of two complementary branches: The first is the BCB, which utilizes a channel attention mechanism to focus on the central regions of the target. The second is the EDCB, which is grounded in a spatial attention mechanism, making it more sensitive to fine details, such as edges and textures. By employing distinct pathways for different feature components, the module effectively mitigates interference between central and boundary pixels, thereby enhancing the model's overall performance.

3.4.1 Body capture branch

The channel attention mechanism autonomously determines the importance of each channel in the input features, amplifying the channel weights for semantic information such as object categories, shapes, and key components, while attenuating the weights for background textures, irrelevant objects, or noise. During the learning process, higher weights are assigned to channels that are more crucial for identifying the primary subject. As shown in Fig. 6a, we first apply global average pooling $\text{GAP}(\cdot)$ and global max pooling $\text{GMP}(\cdot)$ to F_m to integrate global information, obtaining the average pooled feature F_{avg}^c and the maximum pooled feature F_{max}^c . Subsequently, these two features are fed into a MLP for weight learning, and the results are added together to obtain F_{am}^c , thereby improving the joint representation capability of the two features. Finally, to enhance the main subject features, the input feature map is multiplied by the channel weights to obtain the body feature F_{body} , where the weight parameters are obtained through the activation function $\text{Sigmoid}(F_{am}^c)$. The specific process can be expressed as:

$$F_{avg}^c = \text{GAP}(F_m), F_{max}^c = \text{GMP}(F_m), \quad (27)$$

$$F_{am}^c = \text{MLP}(F_{avg}^c) + \text{MLP}(F_{max}^c), \quad (28)$$

$$F_{body} = F_m \otimes \text{Sigmoid}(F_{am}^c). \quad (29)$$

3.4.2 Edge detail capture branch

In the vicinity of object boundaries, adjacent pixel points are most likely to possess similar levels of importance and semantic information. The spatial attention mechanism can highlight object edges or contours by calculating the importance weights of various positions within the feature map. As shown in Fig. 6b, to focus on semantically similar features within spatial neighborhoods and capture detailed information such as textures and edges, we first set up two branches using local average pooling $\text{LAP}(\cdot)$ and local max pooling $\text{LMP}(\cdot)$ to process F_m , obtaining F_{lap} and F_{lmp} . Then, we apply global max pooling and global average pooling separately to F_{lap} and F_{lmp} to aggregate channel information, and concatenate the results, respectively, to achieve better

feature representation capabilities. To minimize information loss caused by multiple pooling operations, a 3×3 convolution is used for further feature learning, obtaining F'_{lap} and F'_{lmp} .

$$F_{lap} = LAP(F_m), \quad F_{lmp} = LMP(F_m), \quad (30)$$

$$F'_{lap} = \text{Conv}(\text{Cat}(\text{GAP}(F_{lap}), \text{GMP}(F_{lap}))), \quad (31)$$

$$F'_{lmp} = \text{Conv}(\text{Cat}(\text{GAP}(F_{lmp}), \text{GMP}(F_{lmp}))). \quad (32)$$

Finally, the processing results of the two branches are multiplied with F_{lap} and F_{lmp} , respectively. And the features are concatenated and deconvolution is performed to restore the feature dimensions, resulting in the detail feature F_{detail} . The specific process can be expressed as:

$$F_{detail} = \text{Conv}_d(\text{Cat}(F'_{lap} \otimes F_{lap}, F'_{lmp} \otimes F_{lmp})). \quad (33)$$

Here, $LAP(\cdot)$ denotes the local average pooling operation, $LMP(\cdot)$ denotes the local max pooling operation and $\text{Conv}_d(\cdot)$ denotes the deconvolution operation.

3.4.3 Feature enhancement

The shallow features of an image retain rich spatial details such as edges and textures due to their high resolution, which are crucial for precise localization and object detection. Introducing low-level features before downsampling can significantly enhance the boundary clarity of segmented targets and alleviate the spatial information degradation problem in deep networks. Therefore, we concatenate the first fused feature F_1^{rt} with F_{body} and F_{detail} along the channel dimension and use the $CBR(\cdot)$ operation for fusion interaction:

$$\begin{cases} F'_{body} = CBR(\text{Cat}(F_{body}, F_1^{rt})), \\ F'_{detail} = CBR(\text{Cat}(F_{detail}, F_1^{rt})). \end{cases} \quad (34)$$

3.5 Loss function

As shown in Fig. 2, the loss function consists of three parts: $\mathcal{L}_{body}$, $\mathcal{L}_{detail}$, and $\mathcal{L}_{gt}$. Based on the label decoding supervision strategy [50], we separate the body ground truth $\mathcal{G}_{body}$ and edge detail ground truth $\mathcal{G}_{detail}$ from the salient object's ground truth $\mathcal{G}$. We use $\mathcal{G}_{body}$ and $\mathcal{G}_{detail}$ to, respectively, supervise the generation of the main body prediction map S_{body} and edge detail prediction map S_{detail} :

$$\mathcal{L}_{body} = \mathcal{L}_{BCE}(S_{body}, \mathcal{G}_{body}), \quad (35)$$

$$\mathcal{L}_{detail} = \mathcal{L}_{BCE}(S_{detail}, \mathcal{G}_{detail}), \quad (36)$$

where $\mathcal{L}_{BCE}$ denotes the binary cross-entropy loss [51].

Meanwhile, under the supervision of the ground truth $\mathcal{G}$, we perform a series of operations including concatenation,

enhancement, and upsampling on the main feature F'_{body} and edge detail feature F'_{detail} . After these operations, we obtain the final significant target prediction map $\mathcal{S}$:

$$\mathcal{L}_{gt} = \mathcal{L}_{BCE}(\mathcal{S}, \mathcal{G}) + \mathcal{L}_{IoU}(\mathcal{S}, \mathcal{G}), \quad (37)$$

where $\mathcal{L}_{IoU}$ indicates the intersection-over-union loss [52].

In summary, the total loss $\mathcal{L}_{total}$ is expressed as:

$$\mathcal{L}_{total} = \mathcal{L}_{body} + \mathcal{L}_{detail} + \mathcal{L}_{gt}. \quad (38)$$

4 Experiments

4.1 Datasets

To comprehensively evaluate the robustness and effectiveness of our proposed SFINet under various complex scenarios, we conducted extensive experiments on three widely recognized benchmark datasets: VT821 [17], VT1000 [53], and VT5000 [54]. Each of these datasets presents unique challenges, providing a rich testing environment for our research. VT821 Dataset: This dataset comprises 821 pairs of RGB-T images from 60 typical scenes, covering a diverse range of target types, quantities, and sizes. The dataset employs manual label alignment, which adds an extra layer of challenge to the saliency detection task. VT1000 Dataset: It includes 1000 pairs of precisely aligned RGB-T images that have undergone rigorous screening and evaluation. These images are accompanied by detailed annotation information, offering a more reliable foundation for model training. VT5000 Dataset: Recognized as one of the largest RGB-T datasets currently available, it contains 5000 image pairs, with 2500 pairs designated for training and the remainder for testing. This dataset not only boasts the most extensive scene library but also covers all-day conditions from day to night, ensuring the model's adaptability under different lighting conditions.

4.2 Evaluation metrics

To compare model performances, we introduce the following evaluation metrics: E-measure (E_m) [55], S-measure (S_m) [56], Weighted F-measure (F_β^ω) [57], F-measure (F_β) [58], MAE ($\mathcal{M}$) [59], and precision-recall (PR) curves. To comprehensively evaluate the experimental results, we employed five widely used metrics. The E-measure evaluates the global similarity and pixel-level correspondence between predictions and ground truth. The S-measure evaluates the concordance of regional characteristics and object structures between the predicted outcomes and the actual ground truths. The F-measure quantifies regional similarity

Table 1 Performance comparisons between proposed SFINet and 11 SOTA methods on the VT821, VT1000 and VT5000 datasets. The arrow $\downarrow$ / $\uparrow$ indicates that the lower/higher value is better. The best two results are highlighted using bolditalic and bold in that order

Method	VT821					VT1000					VT5000				
	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	$F_\beta \uparrow$	$\mathcal{M} \downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	$F_\beta \uparrow$	$\mathcal{M} \downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	$F_\beta \uparrow$	$\mathcal{M} \downarrow$
ADF ₂₃	0.842	0.810	0.627	0.717	0.077	0.921	0.910	0.804	0.847	0.034	0.891	0.864	0.722	0.778	0.048
LSNet ₂₃	0.911	0.878	0.809	0.826	0.033	0.935	0.925	0.887	0.885	0.023	0.915	0.887	0.806	0.825	0.037
HRTransNet ₂₃	0.929	0.906	0.849	0.853	0.026	0.945	0.938	0.913	0.900	0.017	0.945	0.912	0.870	0.871	0.025
CAVER ₂₃	0.929	0.898	0.846	0.854	0.026	0.949	0.938	0.912	0.906	0.016	0.935	0.899	0.849	0.856	0.028
WaveNet ₂₄	0.929	0.912	0.863	0.857	0.024	0.952	0.945	0.921	0.909	0.015	0.940	0.911	0.864	0.866	0.026
LAFB ₂₄	0.915	0.884	0.817	0.843	0.034	0.945	0.932	0.905	0.905	0.018	0.931	0.893	0.841	0.857	0.030
CONTRINet ₂₄	0.926	0.892	0.840	0.850	0.027	0.948	0.929	0.907	0.906	0.019	0.934	0.894	0.848	0.863	0.030
FRPNet ₂₄	0.925	0.895	0.846	0.862	0.029	0.969	0.938	0.921	0.925	0.017	0.938	0.902	0.862	0.880	0.029
MAGNet ₂₄	0.930	0.909	0.863	0.865	0.026	0.949	0.938	0.917	0.913	0.016	0.943	0.909	0.870	0.875	0.025
HENet ₂₅	0.935	0.913	–	0.873	0.025	0.954	0.944	–	0.918	0.016	0.945	0.918	–	0.881	0.024
PLFRNet ₂₅	0.941	0.918	–	0.883	0.022	0.951	0.939	–	0.915	0.015	0.946	0.911	–	0.886	0.024
SFINet (Ours)	0.936	0.916	0.869	0.868	0.022	0.952	0.945	0.921	0.911	0.015	0.948	0.918	0.877	0.880	0.023

using the harmonic mean of precision and recall. Complementing this, the weighted F-measure computes a weighted harmonic mean of precision and recall, prioritizing local pixel neighborhoods. The mean absolute error quantifies the average absolute pixel-wise error across the entire image. Finally, the precision–recall curve provides a comprehensive picture of binary classification performance by plotting precision against recall across all decision thresholds.

4.3 Implementation details

We employ a pre-trained ConvNeXt-B as the backbone architecture of the network and conduct model training on an NVIDIA RTX 3090 GPU. For training, the training subset of the VT5000 dataset is used as the training set, while the VT1000 dataset serves as the validation set. Input images are uniformly resized to a resolution of 384×384 , and data augmentation strategies such as random cropping, horizontal flipping, and multi-angle rotation are applied to enhance the model's generalization capability. During training, a total of 80 epochs are set with a batch size of 4. The optimizer used is stochastic gradient descent (SGD) [60], with momentum and weight decay parameters set to 0.9 and 0.0005, respectively. The maximum learning rate for the backbone network is set to 0.001, while that for the remaining parts is 0.01. The learning rate is decayed every 10 epochs.

4.4 Model comparison

To evaluate the effectiveness and robustness of SFINet, we conduct experiments on three benchmark datasets and compare our results with 11 state-of-the-art (SOTA) RGB-T SOD methods: ADF [54], LSNet [30], HRTransNet [26], CAVER

[38], WaveNet [27], LAFB [61], CONTRINet [32], FRPNet [31], MAGNet [62], HENet [63], and PLFRNet [64]. All competing models are trained on the VT5000 dataset to ensure a consistent training foundation. In the comparative evaluation, we adopt the same performance metrics for all models to guarantee fairness. For those methods where predicted saliency maps are not publicly available, we retrain the models using their open-source code and the training settings described in their original papers to generate the corresponding results.

4.4.1 Quantitative evaluation

As shown in Table 1, we performed a quantitative comparison of the proposed SFINet with 11 other SOTA methods based on five evaluation metrics. The experimental results demonstrate that our model performs equal to or better than HENet and MAGNet in key metrics such as E_m , S_m , F_m , and $\mathcal{M}$. On the VT821 dataset, compared to the HENet, our model achieves improvements of 0.3% in S_m and $\mathcal{M}$. On the VT1000 dataset, compared to the MAGNet, our method shows improvements of 0.7% and 0.4% in S_m and F_m^ω , respectively. The VT5000 dataset, which includes various complex scenes and challenging targets, is one of the most demanding benchmark datasets currently available. Our method outperforms the best model PLFRNet by 0.2% and 0.7% in the E_m and S_m . These results demonstrate the significant advantages of SFINet in prediction accuracy, object structure preservation, and detail enhancement. Furthermore, compared to HRTransNet and WaveNet, our method exhibits superior performance. This can be attributed to our cross-modal fusion strategy, which integrates high-level semantic information with frequency-domain global details. This strat-

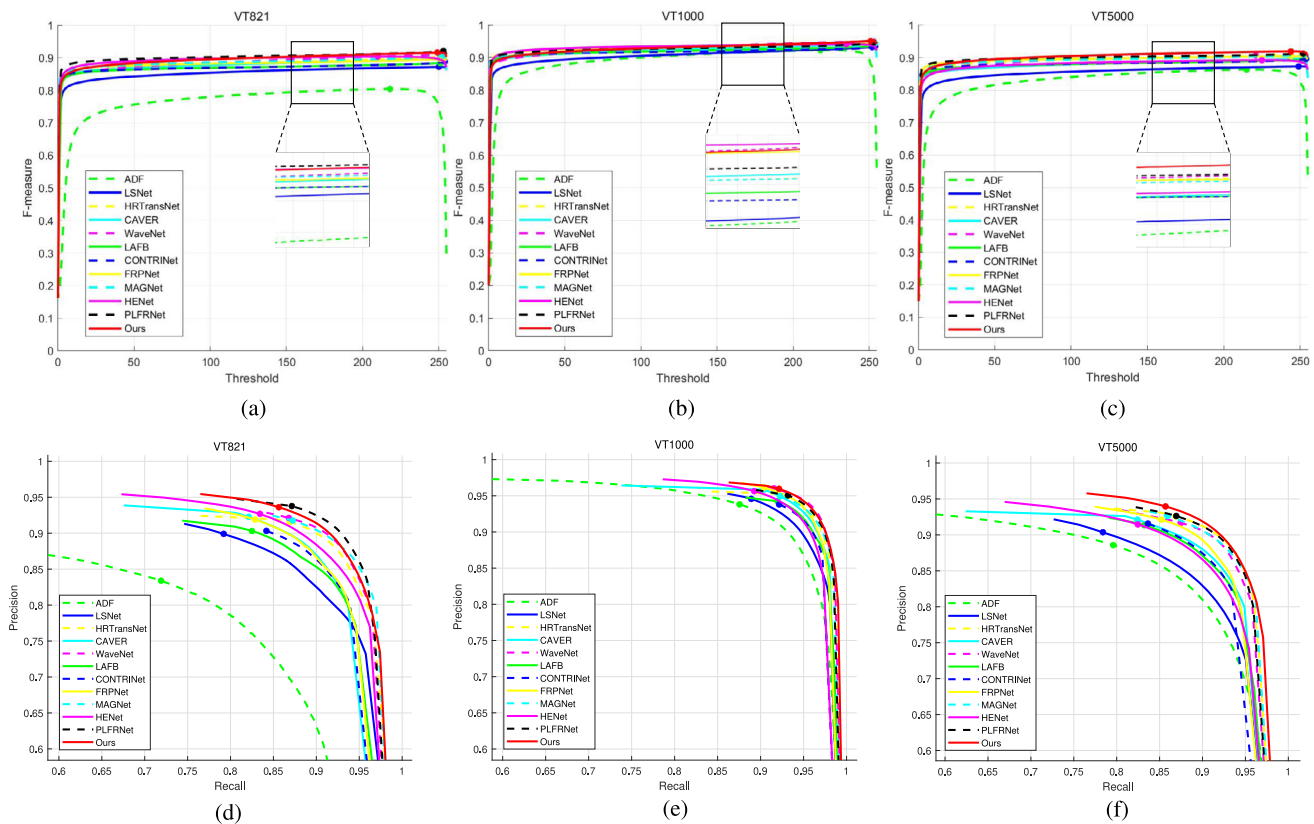


Fig. 7 F-measure curves and precision–recall (PR) curves of our SFINet and 11 SOTA models. **a–c** represent the F-measure curves on the VT821, VT1000, and VT5000 datasets, respectively, while **d–f** show the PR curves on the VT821, VT1000, and VT5000 datasets

egy effectively aligns multi-scale spatial-domain information with frequency-domain data for semantic feature alignment, achieving efficient fusion of complementary information and enhancing the model’s ability to handle complex scenarios. Figure 7 illustrates the F-measure and PR curves. From the F-measure curve, it is evident that our method maintains a significant competitive advantage across different thresholds. When the threshold exceeds 250, the performance of other models declines sharply, whereas our method retains higher precision. Additionally, the comparison of PR curves shows that our curve is closer to the top-right corner, indicating superior performance in both precision and recall.

4.4.2 Qualitative comparison

Figure 8 presents an intuitive comparison between our proposed method and 11 SOTA models across a variety of challenging scenarios. These include complex objects (rows 1–4), intricate backgrounds (rows 4, 5, 6, 10), low-quality thermal infrared images (rows 3, 7, 9, 11), extreme low-light conditions (rows 7, 8), noisy RGB images (row 9), as well as scenes involving small or multiple targets (rows 10, 11). The experimental results demonstrate that even when uni-modal or bimodal information is partially compromised (e.g.,

in low-quality thermal infrared images, extreme low-light environments, noisy RGB images, or when thermal infrared modality detections are missing), our method consistently delivers excellent detection performance. This advantage is primarily attributed to the synergistic mechanism of spatial-domain semantic exploration and frequency-domain global information mining, which enhances the network’s robustness against modality failures. Furthermore, when confronted with low-contrast, multi-scale targets or scenarios involving multiple targets, our method reliably locates targets and achieves precise segmentation. In contrast, existing methods often face challenges such as target omission, blurred boundaries, or redundant responses. This superior performance is largely due to the dual-domain semantic interaction module, which effectively fosters the unification and complementarity of information across different representation domains, thereby improving the consistency and integrity of cross-modal features. Additionally, when dealing with complex object structures or scenes, our method demonstrates enhanced semantic understanding capabilities. For instance, in rows 1 and 2, WaveNet, which relies on global context exploration and frequency-domain analysis, erroneously identifies gaps within the target as foreground regions. In contrast, our approach suppresses noise and irrel-

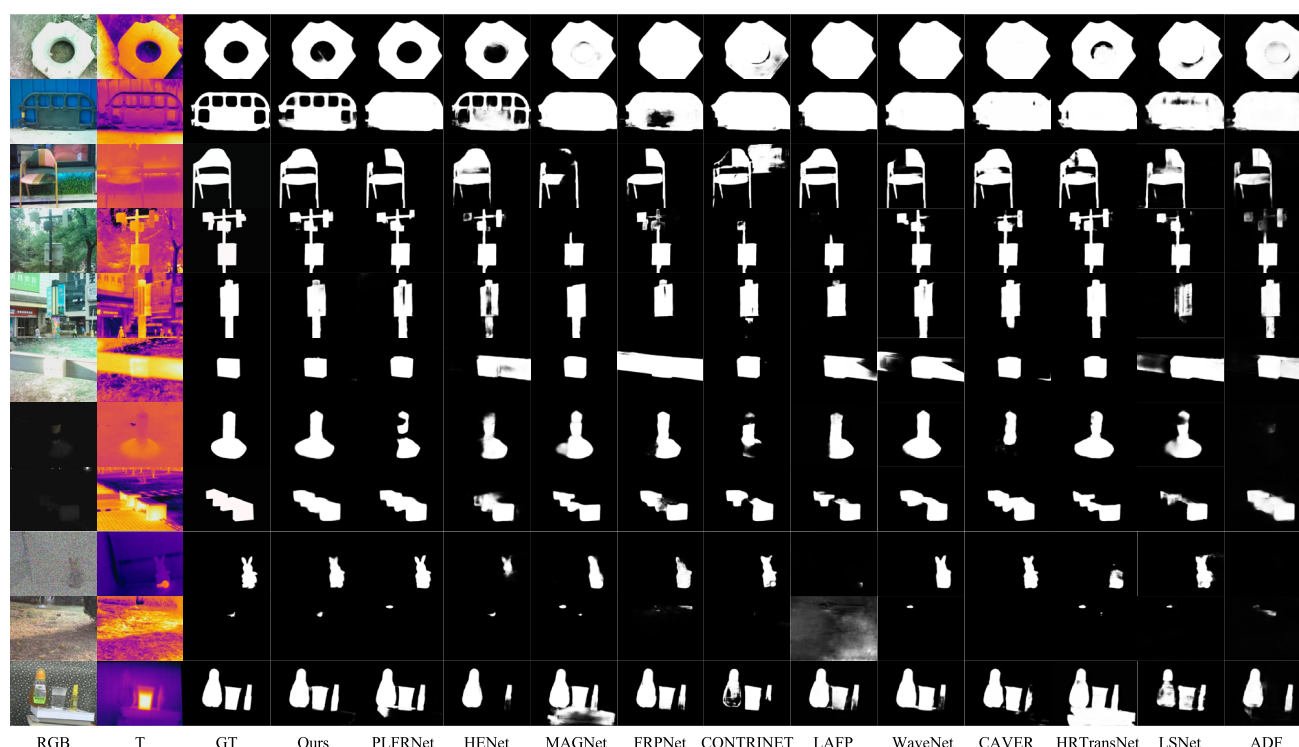


Fig. 8 Visual comparison of our SFINet with 11 SOTA models in various complex conditions

evant texture interference through Fourier transformation and combines decoupled learning mechanisms for main body features and local details. This enables a more accurate perception of the target's overall structure and edge details, resulting in precise boundary segmentation. In summary, the proposed SFINet demonstrates exceptional robustness and generalization capabilities under various challenging conditions, validating its effectiveness and superiority in performing RGB-T salient object detection in complex environments.

4.5 Ablation study

In order to systematically assess the effectiveness of each critical component in SFINet, we conducted a series of ablation studies, emphasizing the validation of the backbone network choice and the contribution of individual functional modules within the network architecture. All experiments were conducted on three benchmark datasets to ensure the comprehensiveness and reliability of the evaluation outcomes. Table 2 and Fig. 9 present the quantitative performance comparison and visual prediction results for various experimental configurations, respectively, where "w/o" denotes model variants with the corresponding component removed.

4.5.1 Effectiveness of backbone

ConvNeXt-B retains the high efficiency of traditional convolutions while achieving an optimal balance between performance and efficiency by incorporating design concepts such as large kernel convolutions and hierarchical structures from Transformers. Its feature processing capability surpasses that of Swin Transformer, leading to its widespread application in various fields including image classification, object detection, and segmentation. Therefore, we chose ConvNeXt-B as the backbone network and conducted a systematic comparison with mainstream backbone networks, including ResNet-50 [65], Swin Transformer [66], and CDF-Former [46]. The model metrics after changing the backbone network are shown in the first three rows of Table 2. When using ResNet-50 and Swin Transformer as the backbone networks, the model's various performance indicators failed to meet the expected results. Additionally, although CDF-Former, based on FFT, possesses strong global context perception and efficient decomposition capabilities for multi-scale features, five key evaluation metrics— E_m , S_m , F_m^ω , F_m , and $\mathcal{M}$ —still experienced average decreases of 1.7%, 1.6%, 2.9%, 2.8%, and 0.6% compared to the original model, respectively. The visualization of prediction results in Fig. 9 further validates this conclusion, demonstrating the superior performance of ConvNeXt-B in complex scenarios.

Table 2 Ablation experiment results of each component in the proposed SFNet. "w/o" means without the module. The best results are marked in bold

Type	Method	VT821			VT1000			VT5000			$\mathcal{M} \downarrow$	$F_\beta \uparrow$	$F_\beta^\omega \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	$F_\beta \uparrow$	$\mathcal{M} \downarrow$
		$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	$F_\beta \uparrow$	$\mathcal{M} \downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	$F_\beta \uparrow$							
Backbone	ResNet-50	0.889	0.864	0.776	0.798	0.039	0.935	0.928	0.891	0.885	0.021	0.908	0.877	0.801	0.815	0.036	
	Swin-B	0.900	0.875	0.798	0.817	0.034	0.943	0.936	0.901	0.895	0.018	0.922	0.887	0.822	0.839	0.031	
Module	CDFFormer	0.921	0.903	0.844	0.841	0.028	0.938	0.935	0.900	0.890	0.019	0.927	0.893	0.836	0.843	0.032	
	w/o SCFA	0.921	0.907	0.845	0.844	0.027	0.945	0.941	0.913	0.903	0.017	0.937	0.909	0.855	0.860	0.027	
	w/o FCPA	0.917	0.901	0.843	0.840	0.026	0.945	0.938	0.909	0.899	0.017	0.938	0.910	0.858	0.858	0.025	
	w/o DSIM	0.923	0.901	0.844	0.850	0.027	0.946	0.936	0.907	0.902	0.017	0.937	0.905	0.852	0.861	0.026	
	w/o FDM	0.915	0.904	0.833	0.837	0.028	0.946	0.940	0.910	0.902	0.017	0.938	0.910	0.856	0.859	0.026	
	w/o BCB	0.924	0.908	0.847	0.848	0.025	0.948	0.940	0.913	0.903	0.016	0.943	0.913	0.866	0.869	0.024	
	w/o EDCB	0.922	0.905	0.843	0.848	0.027	0.946	0.942	0.914	0.904	0.016	0.940	0.909	0.856	0.863	0.025	
	Full model	0.936	0.916	0.869	0.868	0.022	0.952	0.945	0.921	0.911	0.015	0.948	0.918	0.877	0.880	0.023	

4.5.2 Effectiveness of SCFA

To evaluate the impact of the SCFA module on network performance, we adopted an alternative approach: concatenating the input bimodal features along the channel dimension and using a 3×3 convolution for fusion, thereby replacing the SCFA module. As shown in Table 2, the model performance with spa included significantly outperforms the version without this module, especially on the more challenging VT5000 dataset. Additionally, in the w/o SCFA column of Fig. 9, the model lacking the SCFA module exhibits a decline in structural judgment capability, resulting in missing or redundant structures for some targets.

4.5.3 Effectiveness of FCPA

We employed the same validation method used for the SCFA module to test the overall performance improvement brought by the FCPA module. The experimental results are shown in the w/o FCPA group of Table 2. Compared to the complete model, the model without the FCPA module shows varying degrees of performance degradation across three datasets. From the w/o FCPA column of Fig. 9, it can be seen that introducing global information processing in the frequency domain effectively reduces noise interference in the prediction map while enhancing the integrity of target structures.

4.5.4 Effectiveness of DSIM

To validate the effectiveness of fusion, we chose the same verification method as for the SCFA module to process spatial-domain and frequency-domain features. As indicated in the w/o DSIM group of Table 2, the performance of this group is noticeably lower than that of the complete model, with average reductions of 2.1% and 1.6% in F_m^ω and F_m , respectively, across the three datasets. Furthermore, in the prediction map comparison of Fig. 9, the model lacking the DSIM module struggles with understanding scenes involving multiple targets or targets composed of multiple parts, leading to issues such as missing targets or incomplete structures.

4.5.5 Effectiveness of FDM

To thoroughly validate the effectiveness of the FDM, we designed three ablation experiments: w/o FDM denotes the removal of the FDM along with its corresponding main loss and detail loss; w/o BCB indicates the removal of the BCB and the main loss; w/o EDCB refers to the removal of the EDCB and the detail loss. As shown in Table 2, the FDM led to an average degradation of 1.23%, 0.83%, 2.27%, 2.03%, and 0.37% in E_m , S_m , F_m^ω , F_m , and $\mathcal{M}$, respectively. Furthermore, ablating either branch also resulted in a noticeable

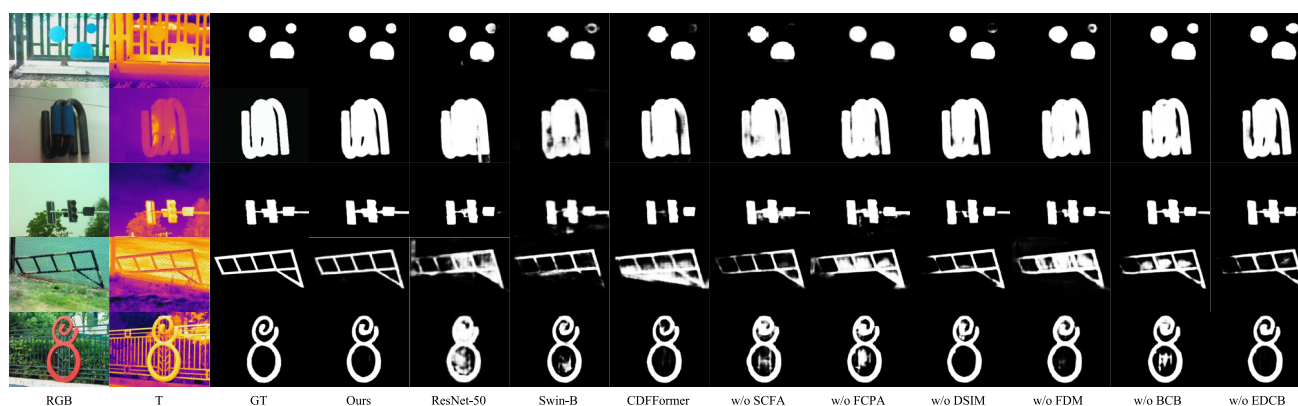


Fig. 9 Visualization results of ablation experiments

Table 3 Quantitative Comparison of Different Approaches Regarding Parameters and FLOPs

Method	Backbone	Params/M	FLOPs/G
WaveNet	Wave-MLP	64.0	80.7
LAFB	Res2Net	453.03	139.73
CONTRINet	Res2Net50	34.77	37.12
MAGNet	MobileNetV2	16.1	9.9
HENet	MobileViT-V1	11.92	10.75
PLFRNet	PVTv2	41.9	25.1
SFINet(Ours)	ConvNext-B	98.6	123.96
Ours	CDFFormer	38.3	57.08
Ours	Swin-B	97.9	123.66

performance decline across all metrics. From the visualization results in the fourth row of Fig. 9, it can be observed that the w/o FDM group yields notably inferior predictions, the w/o BCB group contains some misjudged regions, and the w/o EDCB group suffers from incomplete edge structures. These results demonstrate that the FDM plays a critical role in enhancing model performance, and the two branches are able to achieve effective information complementarity.

4.6 Complexity analysis

As shown in Table 3, a comparison between SFINet and other SOTA methods is presented in terms of parameter count and FLOPs. We selected ConvNeXt-B as the backbone network, which contains 98.6M parameters and 123.96G FLOPs. By experimenting with different backbones, it is observed that the computational cost of SFINet is primarily concentrated in the feature extraction backbone. Compared to other SOTA methods, our approach does not exhibit a significant advantage in terms of parameter count and FLOPs, making it difficult to meet real-time requirements. Consequently, this technology is challenging to deploy on low-power devices. Therefore, introducing lightweight backbone networks or

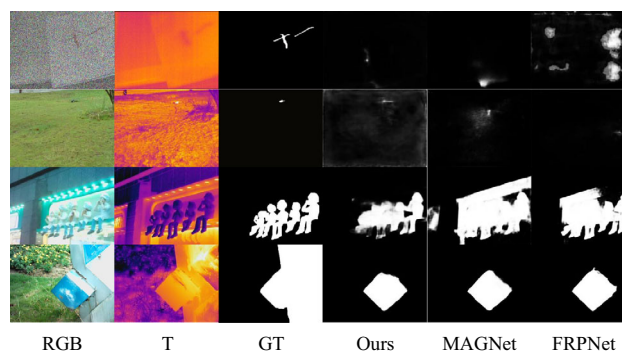


Fig. 10 Failure cases of the proposed SFINet

knowledge distillation [67] techniques constitutes a critical direction for future model refinement, aimed at expanding its scope for practical deployment.

4.7 Failure cases

Despite the promising performance of SFINet across three benchmark datasets, detection failures still occur in highly challenging scenarios, as demonstrated in Fig. 10. In the first example of the top row, severe degradation of image quality from both modalities impairs the model's ability to extract relevant information about salient objects, leading to detection failure. This issue is not unique to the proposed method, as other advanced models also face similar challenges under these conditions. In the second example, small salient objects embedded within a texture-rich grassy background, which share semantic similarities with the surrounding environment, result in missed detections or incomplete segmentations across all three methods. The third example illustrates a complex multi-object scenario with intricate boundaries, where background light interference distorts spatial structures. Although the thermal infrared modality provides supplementary information, it still significantly interferes with model predictions. Lastly, in the fourth

example, the salient object consists of a prominent central region and part of the right wall. Since targets in the training data are mostly concentrated in the central part of the image, models fail to learn sufficiently about edge-located targets, resulting in partial recognition of the central part and the omission of the right wall in all three methods. The analysis above highlights the limitations of current models in handling extreme image degradation, small target embedding, complex background interference, and non-centrally distributed targets. Future research could explore advanced feature filtering and selection mechanisms to enhance the model's semantic understanding of complex targets. Meanwhile, considering the introduction of feature domain adaptation methods [68, 69] could further improve the effectiveness of cross-domain fusion and the generalization performance of the model.

5 Conclusion

In this paper, we propose SFINet, an efficient spatial-frequency integration network for RGB-T salient object detection, aiming to mitigate cross-modal discrepancies and achieve complementary information fusion across modalities and domains. Specifically, to effectively fuse dual-modal features, we design a spatial-domain cross-modal fusion attention (SCFA) module and a frequency-domain cross-modal perception attention (FCPA) module from both spatial and frequency perspectives. These modules alleviate modality differences caused by distinct physical properties through long-range dependency modeling and global information integration. Furthermore, to fully integrate features from different domains and reduce semantic gaps, we introduce a dual-domain semantic interaction module (DSIM) to facilitate information exchange across domains. Additionally, a feature decoupling module (FDM) is employed to extract structural content and edge details from the fused features, thereby optimizing loss supervision and improving detection accuracy. Extensive experiments on three benchmark datasets demonstrate that SFINet achieves competitive performance compared to 11 SOTA models and excels particularly in complex scenarios, validating the effectiveness of integrating spatial and frequency-domain information.

Author Contributions Xiaosheng Yu and Jiawei Huang conceived the study and designed the research framework. Xiaosheng Yu performed the data analysis and wrote the initial draft of the manuscript. Jiawei Huang contributed to the methodology development and conducted experiments. Ying Wang assisted with data collection, visualization. Jubo Chen supervised the project, provided critical feedback, and revised the manuscript for important intellectual content. All authors reviewed and approved the final version of the manuscript.

Data Availability The datasets VT821, VT1000 and VT5000 for this study can be found in the <http://chenglongli.cn/code-dataset/>.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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